

Sankalp S. Sonar, M.Sc., MS

Lead Forward Deployment Engineer | AI & Agentic Solutions | 11 Years Experience

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PROFESSIONAL SUMMARY

Lead Forward Deployment Engineer / AI delivery leader with 11 years of experience translating complex business problems into production-ready AI, GenAI, analytics, and data solutions across revenue forecasting, ESG, marketing intelligence, CX, healthcare, and financial services. Hands-on across Python, FastAPI, Dash, LangChain, LangGraph, LlamaParse, OpenAI APIs, vector databases, GCP/AWS, microservices, and enterprise data pipelines. Proven in owning solution architecture from discovery and prototyping through deployment, leading engineering teams, and turning model outputs into explainable, actionable decision intelligence for business stakeholders.

WORK EXPERIENCE

Associate Manager

08/2025 - Present

Sutherland Global, Navi Mumbai, India

Stack: Python (FastAPI, Hugging Face, Pandas, NumPy, Dash, LangChain), LangGraph, LlamaParse, OpenAI APIs, PGVector, Postgres, GCP, Agent Platform, Agent Garden, Model Garden

- Spearheaded the design and deployment of an organization-level ML-driven Revenue Forecasting solution using Dash as the analytical frontend and microservice-enabled FastAPI backend services, with Salesforce dump ingestion for monthly and annual insight generation.
- Owned the end-to-end solution lifecycle spanning data ingestion, analytical processing, service integration, dashboarding, deployment, and stakeholder-facing insight delivery.
- Led a 3-member team in an institutional hackathon, managing task allocation, code reviews, integration, and deployment of **OSIRIS** - a scalable forecast insight agent for comparative analysis and natural-language interpretation, reusable across forecast-generation workflows.
- Currently architecting **AI Marketing Studio**, a GCP cloud-first multi-agent platform combining omnichannel marketing data, predictive analytics, and autonomous recommendations for campaign planning, optimization, attribution, and executive reporting.
- Currently architecting **CX360 Decision Intelligence**, an agentic decision-support solution delivering contextual KPI recommendations across Executive, Manager, and frontline Agent strata to drive continuous performance improvement.
- Leveraging GCP Agent Platform, Agent Garden, Model Garden, LangGraph, LangChain, OpenAI APIs, PGVector, and Postgres to design enterprise-ready agentic solutions and reusable deployment patterns.

AI Team Lead

08/2024 - 08/2025

Powerweave Software Services Pvt. Ltd., Mumbai, India

Stack: Python (LangChain, FastAPI, Hugging Face, Pandas, NumPy), LlamaParse, OpenAI APIs, PGVector/FAISS, AWS EC2, Docker, Bitbucket, Postgres

- Spearheaded AI-driven backend APIs for ESG reporting frameworks (BRSR, GRI) using custom RAG pipelines, LlamaParse-based document ingestion, and LangChain orchestration.
- Led a 3-member AI engineering team, overseeing task delegation, code reviews, engineering quality, and containerized deployment of FastAPI services on AWS EC2.
- Architected high-accuracy retrieval pipelines using PGVector/FAISS and OpenAI-powered LLMs, optimizing semantic search across complex PDF layouts and structured Excel files.
- Standardized ingestion, embedding generation, and retrieval workflows to automate ESG questionnaire population and improve reporting efficiency.
- Delivered both a monolithic API proof of concept and a microservice-enabled MVP for the ESG governance suggestion engine.

Data Team Lead (R-Package Deployment)

07/2023 - 01/2024

Argus India Price Reporting Services LLP (Argus Media Ltd.), Mumbai, India

Stack: R/R-Shiny (Golem, roxygen2, Teal, Plotly, ggplot2), CSS, JavaScript, SQL, MySQL, JIRA, GitHub, Git

- Led data-driven initiatives to optimize operational workflows through R package development, R-Shiny applications, documentation, and reusable analytical components.
- Advanced Argus processes through version-controlled development, data engineering, data management, and team-level technical leadership.

Jr. Data Scientist

11/2022 - 06/2023

Blenheim Chalcot, Mumbai, India

Stack: AWS, Python (PySpark, GPTIndex, LangChain, Pinecone), Node.js, HTML, CSS, JavaScript, Postgres, JIRA

- Designed exploratory data analysis workflows with PySpark for large-scale datasets and explored GenAI/NLP solutions using GPTIndex, LangChain, and Pinecone.
- Supervised data ETL and evaluated big-data alternatives using Hadoop/HiveQL while delivering client dashboards and analytical workflows.
- Implemented Tableau/Tableau Prep and Power BI solutions supporting product launches and data-driven B2B lending decisions.

Centralized Monitor (Analytics)

07/2021 - 11/2022

IQVIA RDS Ltd., Mumbai, India

Stack: Python (PyInstaller, sklearn, SciPy, Matplotlib, Plotly, Dash), GitHub, TIBCO Spotfire

- Developed automation pipelines for Risk-Based Monitoring studies and ML-based outlier detection workflows, presenting results through interactive analytics applications.
- Developed Data Analytics Plans for RBM studies and presented technical findings to key sponsor stakeholders using R, Plotly, Dash, and R-Shiny.
- Reviewed reports from multiple EDC sources under ICH-GCP guidelines and developed reusable R packages to improve workflow reproducibility and analysis efficiency.

Data Scientist (Consultant) <i>Freelance, Mumbai, India</i> Stack: Python (sklearn, SciPy, Matplotlib, Plotly, Dash), Power BI, GitHub	04/2020 - 05/2021
- Performed statistical market-trend analysis supporting intra-day trading insights with 95% accuracy in a Power BI dashboard. - Maintained and optimized R/Python analytical workflows, improving prediction-model workflow efficiency by 10% and presenting findings through Dash.	
Data Scientist (Associate Researcher-II) <i>Mount Sinai Health System, New York, USA</i> Stack: HPC (Minerva), Nextflow, MATLAB, R/R-Shiny, Python, PySpark, TensorFlow, PyTorch, Tableau, Power BI, SQL/MySQL, GitHub	06/2016 - 01/2020
- Developed and optimized RNA-seq and multi-omic data pipelines using R, Nextflow, HPC, reusable package frameworks, testing, profiling, and benchmarking. - Built Python/Dash analytical workflows and transformed large drug and adverse-event databases using NLP, SQL, MySQL, and Tableau. - Designed neural time-series analysis pipelines using PySpark for processing and TensorFlow/PyTorch for model development, with findings visualized in Power BI.	
Database Management Specialist <i>SRL Diagnostics, Mumbai, India</i> Stack: Excel, Advanced VBA, SQL, SAS v9.1.4	05/2013 - 07/2014
Coordinated cancer clinical-trial data activities with clients and managed project data on a commercial LIMS platform.	

SKILLS AND STACK

- Forward Deployment & Leadership:** Solution discovery, business-to-technical translation, rapid prototyping, solution architecture, productionization, stakeholder communication, team leadership, task delegation, code reviews, and reusable deployment design.
- Generative & Agentic AI:** LLMs, RAG, LangChain, LangGraph, LlamaParse, OpenAI APIs, Hugging Face, multi-agent frameworks, semantic search, prompt/context engineering, natural-language insight generation, and autonomous recommendation systems.
- Python, APIs & Application Engineering:** Python, FastAPI, Dash, Pandas, NumPy, SciPy, scikit-learn, REST APIs, monolithic and microservice architectures, Docker, model/service integration, and production API development.
- Cloud & AI Platforms:** GCP, Agent Platform, Agent Garden, Model Garden, AWS EC2, S3, Lambda, SageMaker, containerized services, and cloud-first AI solution architecture.
- Retrieval, Databases & Data Pipelines:** PGVector, Postgres, FAISS, Pinecone, Chroma, SQL, MySQL, Salesforce ingestion, embeddings, semantic retrieval, structured/unstructured document processing, and ETL pipelines.
- Machine Learning & Data Engineering:** PySpark, Hadoop, HiveQL, TensorFlow, PyTorch, forecasting, classification, regression, NLP, statistical modeling, feature engineering, and large-scale data processing.
- Analytics & Visualization:** Dash, Plotly, Tableau, Power BI, R-Shiny, TIBCO Spotfire, executive dashboards, comparative analysis, KPI monitoring, and analytical storytelling.
- Engineering Tooling:** Git, GitHub, Bitbucket, JIRA, R package development, bash scripting, Java, MATLAB, testing, profiling, benchmarking, and reproducible code workflows.

EDUCATION

Executive PG Programme in Data Science - Data Engineering <i>International Institute of Information Technology, Bangalore, India</i> 3.70/4.00 GPA Scale	Oct' 2021 - Nov' 2022
Master of Science in Biomedical Science - Neuroscience <i>Icahn School of Medicine, Mount Sinai, NY</i> 3.65/4.00 GPA Scale	Aug' 2014 - Aug' 2016
Master of Science - Applied Genetics <i>Brindavan College, Bangalore University, Bangalore, India</i> 73.45% or 3.97/4.00 GPA Scale (As per WES Evaluation)	Aug' 2009 - June' 2011
Bachelor of Science - Microbiology <i>Royal College of Science, Mumbai University, Mumbai, India</i> 65.1% or 3.39/4.00 GPA Scale (As per WES Evaluation)	May' 2006 - July' 2009

CERTIFICATIONS

- Google Generative AI Leadership Certification**
Ongoing
- DS C30 Bootcamp Certificate - upGrad**
Issued May 2021 | Credential ID 33211416